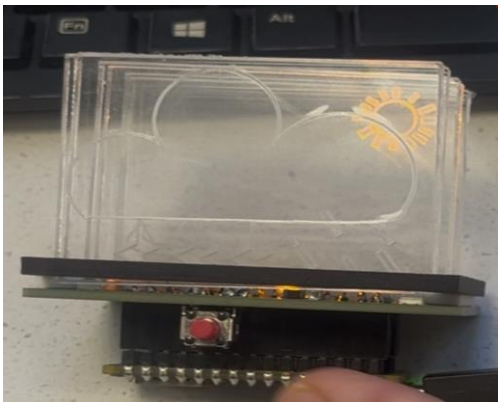
Datasheet –
Modular Weather Display

Product name: Modular Weather Display

Features:

- Low power consumption
- Acrylic display
- Real time weather updates
- WIFI connectivity



Description:

The weather display is a compact, ESP32-based weather display device that retrieves real-time meteorological data via the OpenWeatherMap API over Wi-Fi and maps current weather conditions to a custom LED array mounted on a PCB. Designed for instant at-a-glance weather awareness, Nimbus requires no screen or user interaction to display live local conditions.

Datasheet –
Modular Weather Display

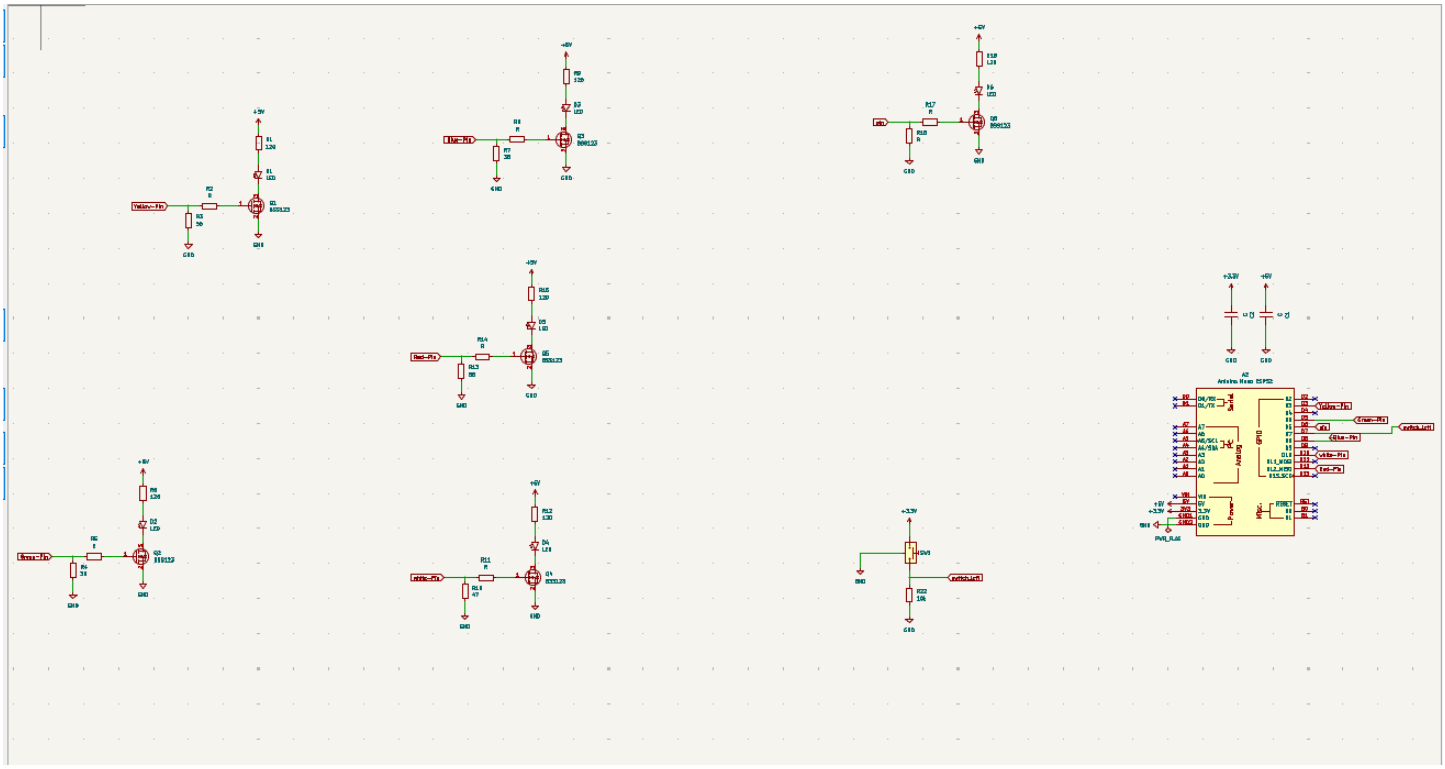
Electrical specifications:

Operating Voltage	3.3v,5v
Current during measurements	150 –250mA
Idle Current	80mA
Maximum power	0.75 W

LED specifications:

LED	**Foward voltage**	**Current**
Blue x2	3.2v	15mA
Yellow x2	2.1v	24mA
Green	3.2v	15mA
Red	2.1v	24mA

PCB Schematic



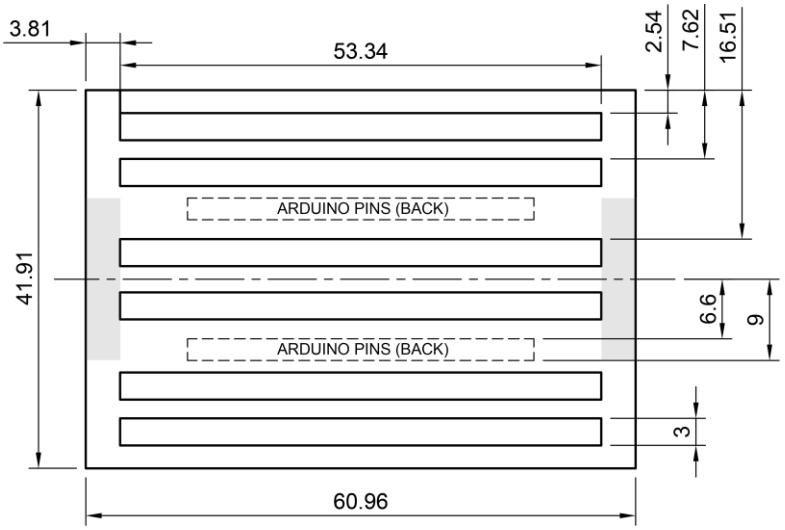
PINOUT:

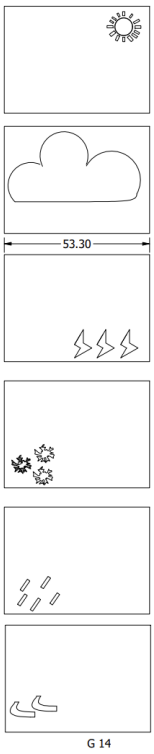
Pin	Function
D0	
D1	
D2	
D3	Yellow LED 1
D4	
D5	Green LED
D6	Yellow LED 2
D7	Button
D8	Blue LED 1
D9	
D10	Blue LED
D11	
D12	Red LED
D13	
A0	
A1	
A2	
A3	
A4	
A5	
A6	
A7	
Vin	
5V	+5V_USB
3.3V	+3.3V

Datasheet –
Modular Weather Display

Mechanical Design :

41.96 x 60.96 x 53.3 mm





DRAWN jmfm22 yv242 da2144	20/04/2026		
CHECKED		TITLE	
QA		Project Week 4 Acrylic Sheets	
MFG			
APPROVED			
		SIZE A3	DWG NO Drawing2 REV
		SCALE	SHEET 1 OF 1